1. Single-Area OSPF
2. Multi area OSPF
3. OSPF Tuning and Troubleshooting

Instructor Materials

Connecting Networks v6.0

1. Single-Area OSPF

2. Multiarea OSPF

3. OSPF Tuning and Troubleshooting

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Single-Area OSPF

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- **It was created to give instructors a PowerPoint to take and modify as their own.**
- **This PowerPoint is:**

  NOT a study guide for the module final assessment.

  NOT a study guide for the CCNA certification exam.
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Objectives

- **Link-state routing protocol**
- **Single-area OSPF concepts**
- **Single-area OSPF configuration**

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Overview of Link-State and Distance Vector Routing

Protocol	Description	Characteristics
Distance-Vector Protocols	RIP V1 & RIP V2 Interior Gateway Routing Protocol (IGRP)	• Copies routing table to neighbors • Updates frequently • RIP V1 and RIP V2 use hop count as metric • Views network from neighbors' perspective • Slow to converge • Susceptible to routing loops • Easy to configure and administer • Consumes quite a bit of bandwidth
Link-State Protocols	Open Shortest Path First (OSPF) Intermediate-System to Intermediate-System (IS-IS)	• Uses shortest path • Updates via event triggers • Sends link-state packets to all network routers • Has common view of network • Fast to converge • Not as susceptible to routing loops • Harder to configure • Requires more memory and processing power than DV • Consumes less bandwidth than DV

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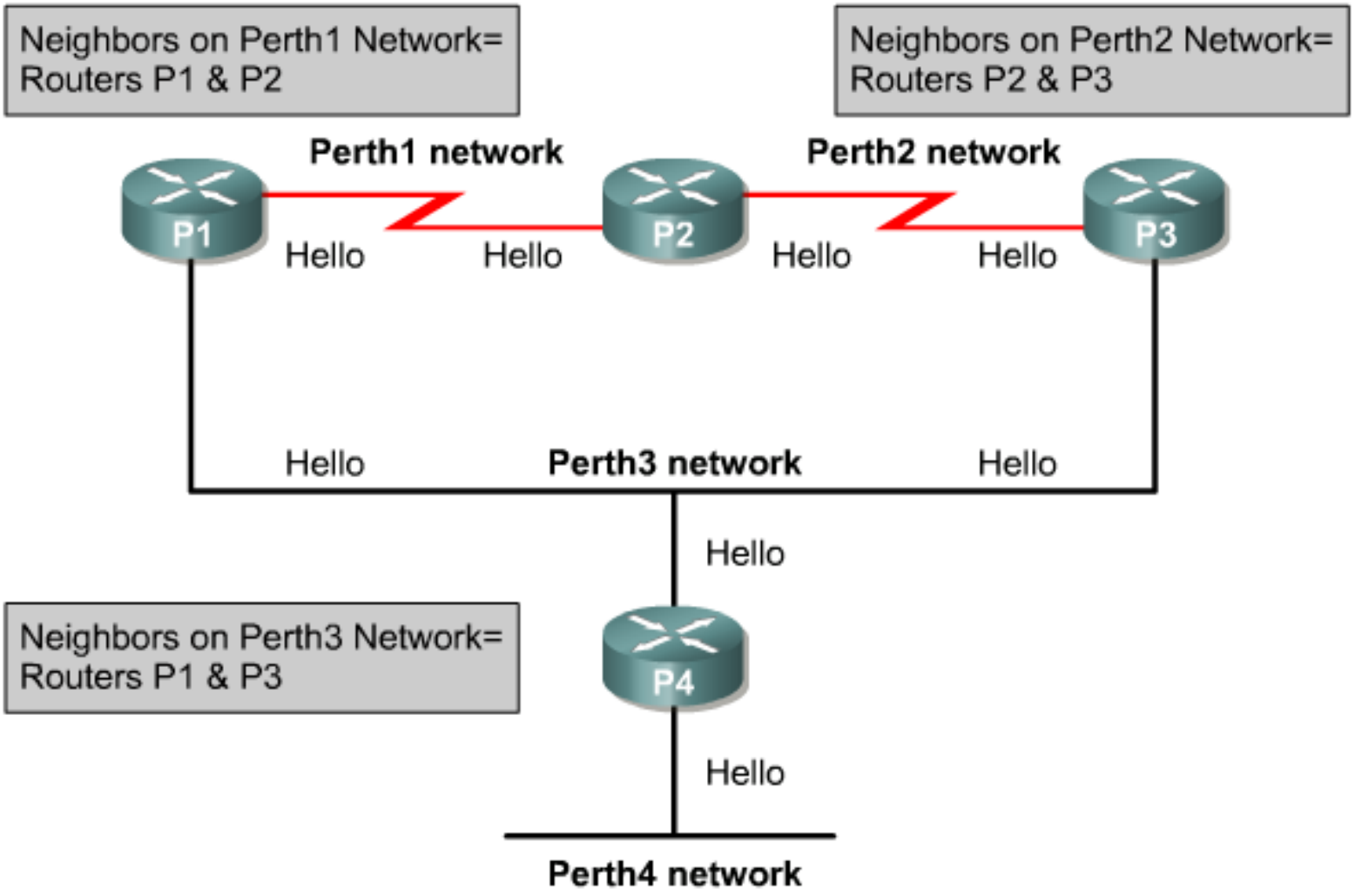
Link-State Routing Features



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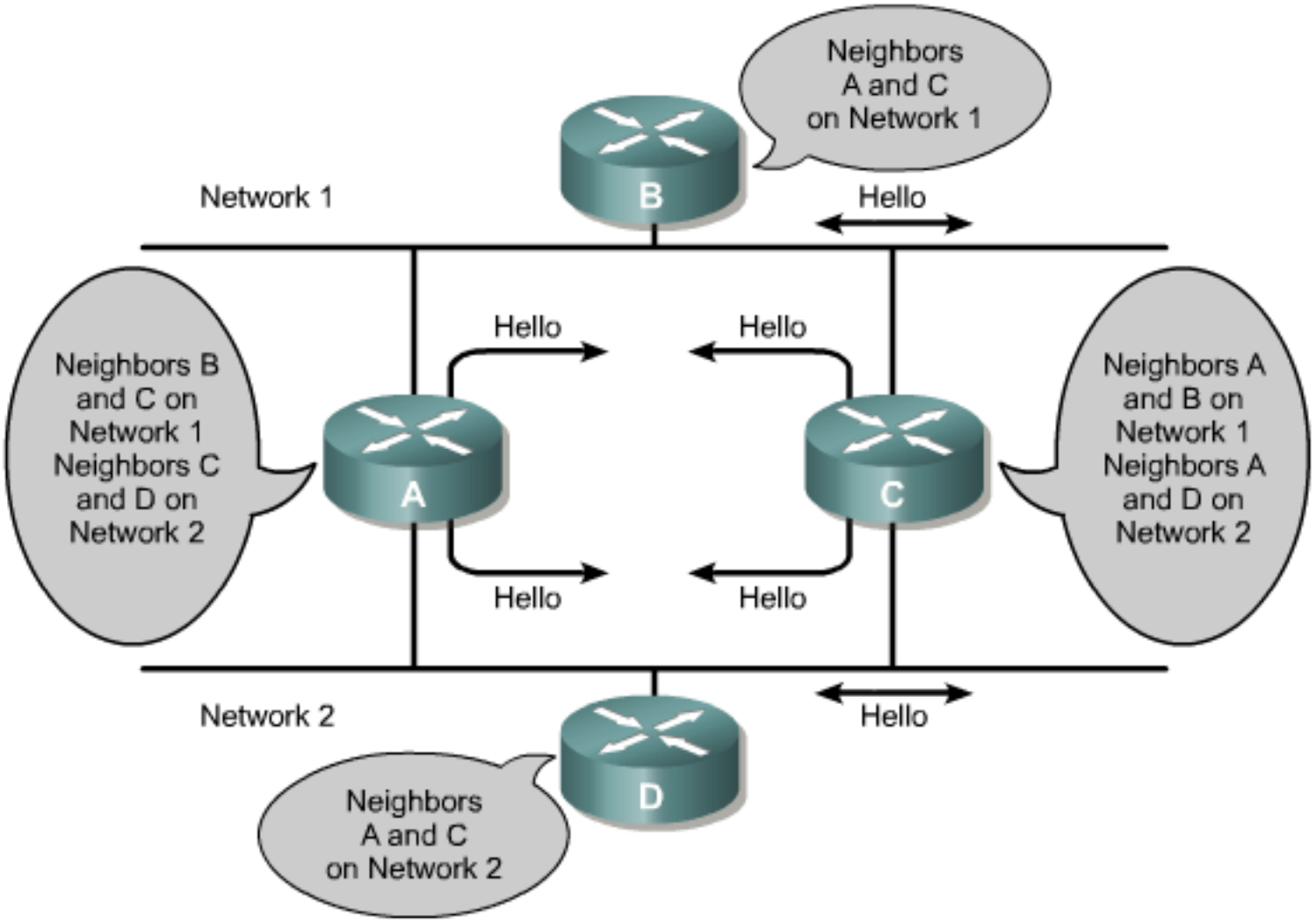
How Routing Information Is Maintained



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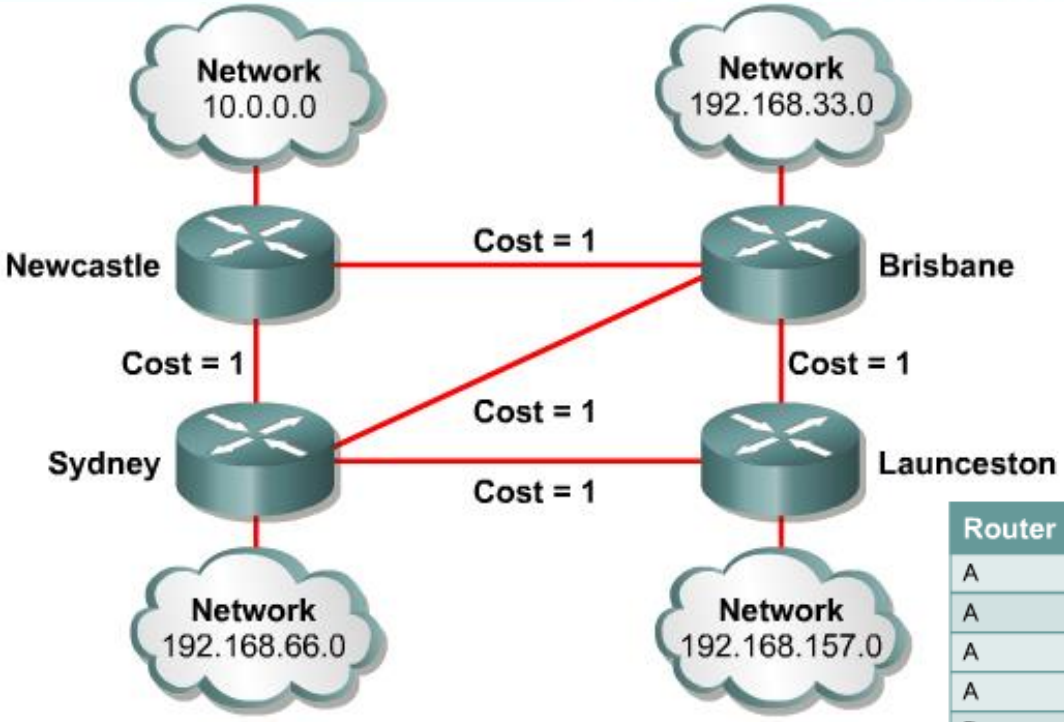
Link-State Routing Protocol Algorithms



Router	Destination	Next Hop	Cost
A	192.168.66.0	B	1
A	192.168.33.0	C	1
A	192.168.157.0	B	2
A	192.168.157.0	C	2
B	10.0.0.0	A	1
B	192.168.33.0	C	1
B	192.168.157.0	D	1
C	10.0.0.0	A	1
C	185.134.0.0	B	1
C	192.168.157.0	D	1
D	10.0.0.0	B	2
D	10.0.0.0	C	2
D	192.168.66.0	B	1
D	192.168.33.0	C	1

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Advantages and Disadvantages of Link-State Routing

Advantages	Disadvantages
• Fast convergence: changes are reported immediately by the source affected • Robustness against routing loops • Routers know the topology • Link-state packets are sequenced and acknowledged • The link-state database sizes can be minimized with careful network design	• Significant demands on memory and processing resources • Requires very strict network design • Requires a knowledgeable network administrator • Initial flooding can impede network performance

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Comparing Distance Vector and Link-State Routing

Distance Vector	Link State
• View network topology from neighbor's perspective • Adds distance vectors from router to router • Frequent, periodic updates: Slow convergence • Passes copies of routing tables to neighbor routers	• Gets common view of entire network topology • Calculates the shortest path to other routers • Event-triggered updates: Faster convergence • Passes link-state routing updates to other routers

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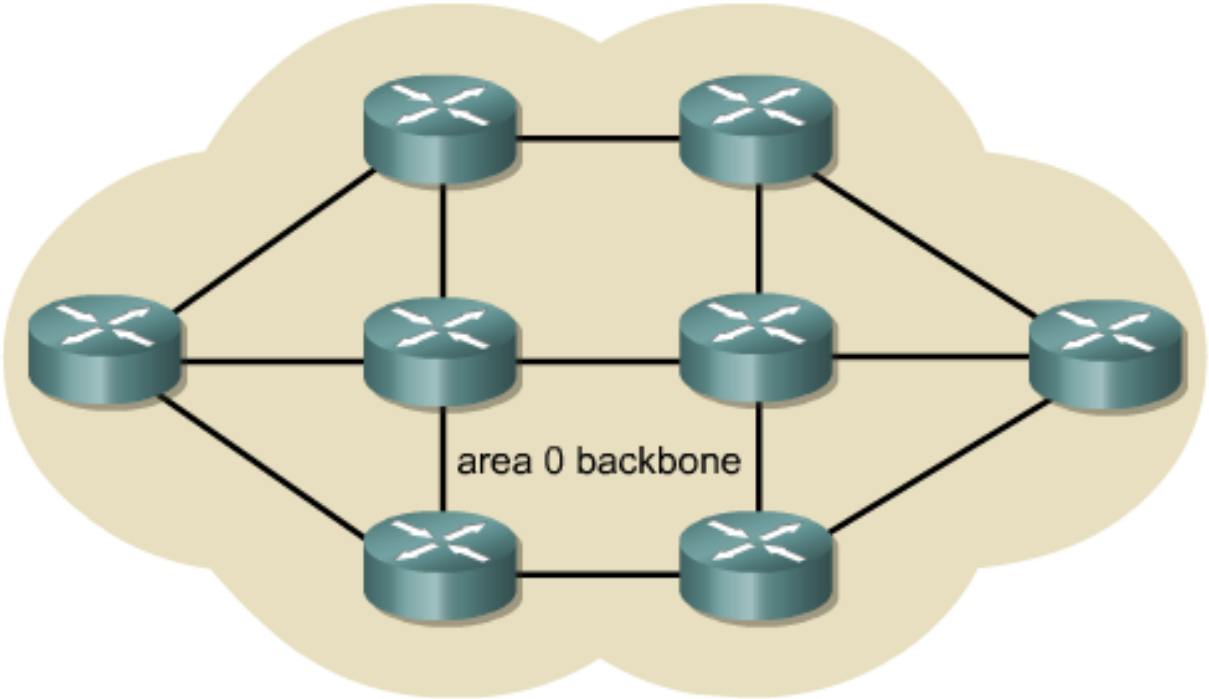
OSPF Overview



OSPF is becoming the preferred IGP protocol when compared with RIPv1 and RIPv2 because it is scalable.

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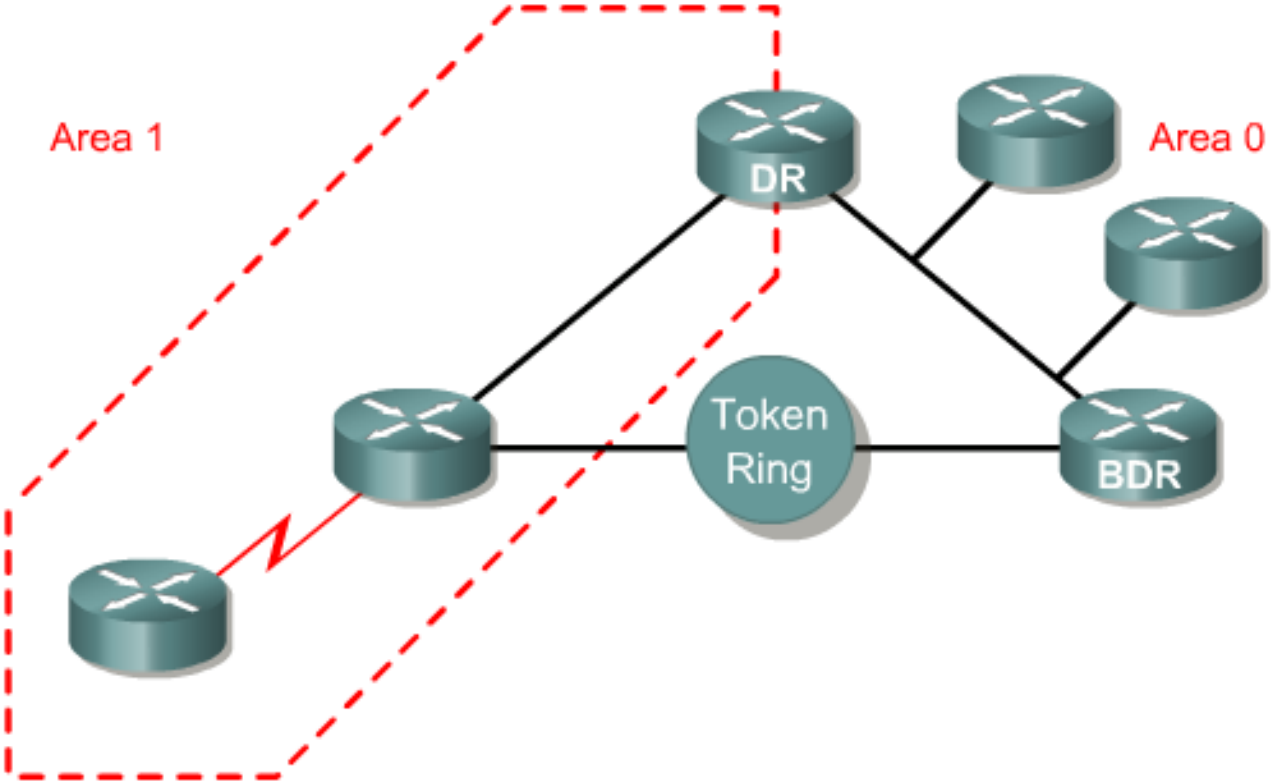
OSPF Terminology



Area: A collection of networks and routers that has the same area identification. Each router within an area has the same link-state information. A router within an area is called an internal router.

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More OSPF Terminology

Link: An interface on a router.

Link-State: The status of a link between two routers. Also a router interface and its relationship to its neighboring routers.

Link-state database (or topological database): A list of information about all other routers in the internetwork. It shows the internetwork topology.

Cost: The value assigned to a link. Rather than hops, link-state protocols assign a cost to a link, which is based on the bandwidth of the link (transmission speed).

Routing table: The routing table (also known as forwarding database) generated when an algorithm is run on the link-state database. Each router's routing table is unique.

Adjacencies database: A listing of all the neighbors to which a router has established bidirectional communication.

Designated router (DR) and backup designated router (BDR): A router that is elected by all other routers on the same LAN to represent all the routers. Each network has a DR and BDR.

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Comparing OSPF Link State with Distance Vector Routing Protocols

Distance Vector	Link State
• View network topology from neighbors perspective • Adds distance vectors from router to router • Frequent, periodic updates: Slow convergence • Passes copies of routing tables to neighbor routers • Use flat topology	• Gets common view of entire network topology • Calculates the shortest path to other routers • Event-triggered updates: Faster convergence • Passes link-state routing updates to other routers • Allows hierarchical design for large internetworks

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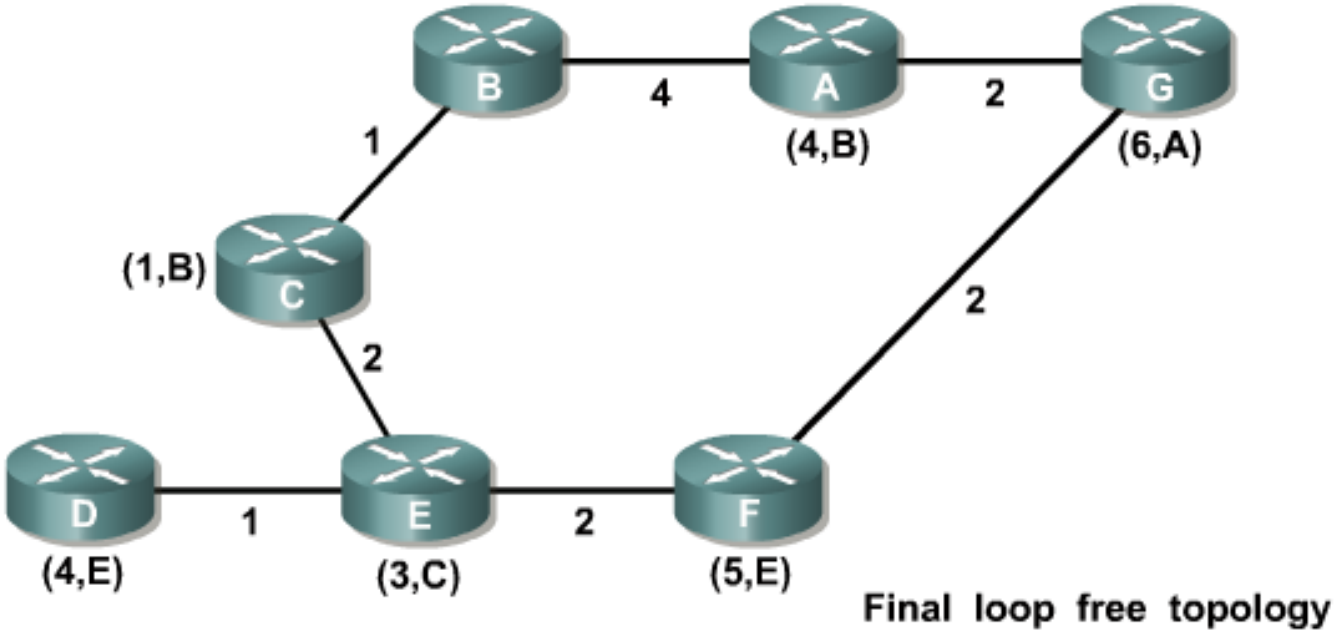
Shortest Path Algorithm



A	B	C	D	E	F	G
B/4	A/4	B/1	C/4	C/2	E/2	A/2
G/2	C/1	D/4	E/1	D/1	G/2	F/2
		E/2		F/2		

The best path is the lowest-cost path.

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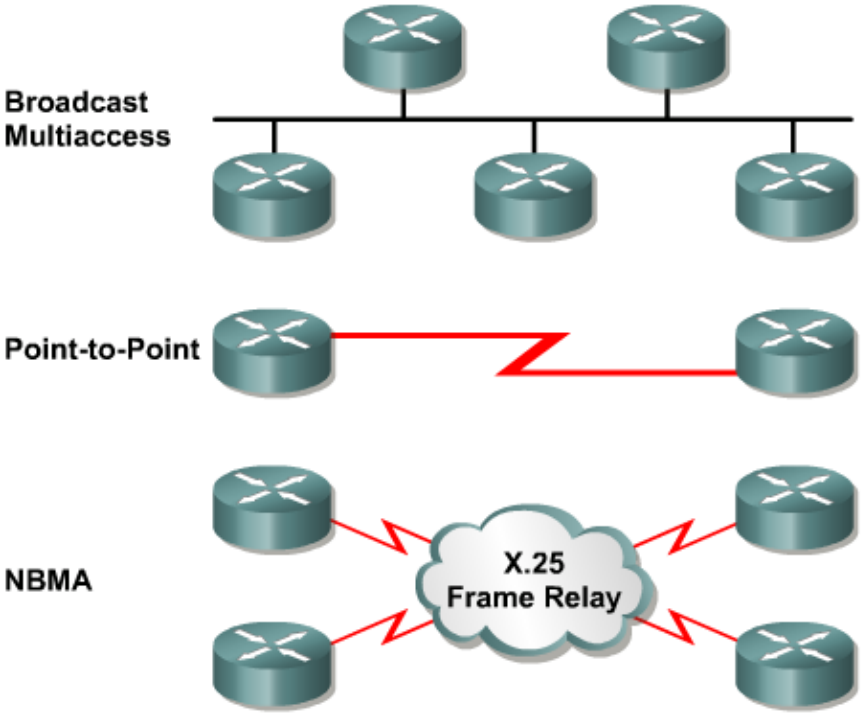
OSPF Network Types



Network Type	Characteristics	DR Election?
Broadcast multiaccess	Ethernet, Token Ring, or FDDI	Yes
Nonbroadcast multiaccess	Frame Relay, X.25, SMDS	Yes
Point-to-point	PPP, HDLC	No
Point-to-multipoint	Configured by an administrator	No

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OSPF Hello Protocol

Version	Type	Packet Length
Router ID		
Area ID		
Checksum		Authentication Type
Authentication Data		

Network Mask		
Hello Interval	Options	Router Priority
Dead Interval		
Designated Router		
Backup Designated Router		
Neighbor Router ID		
Neighbor Router ID		
(additional Neighbor Router ID fields can be added to the end of the header, if necessary)		

Hello packet carries information about hello and dead interval times router identifications. Routers must agree about this information to form adjacencies.

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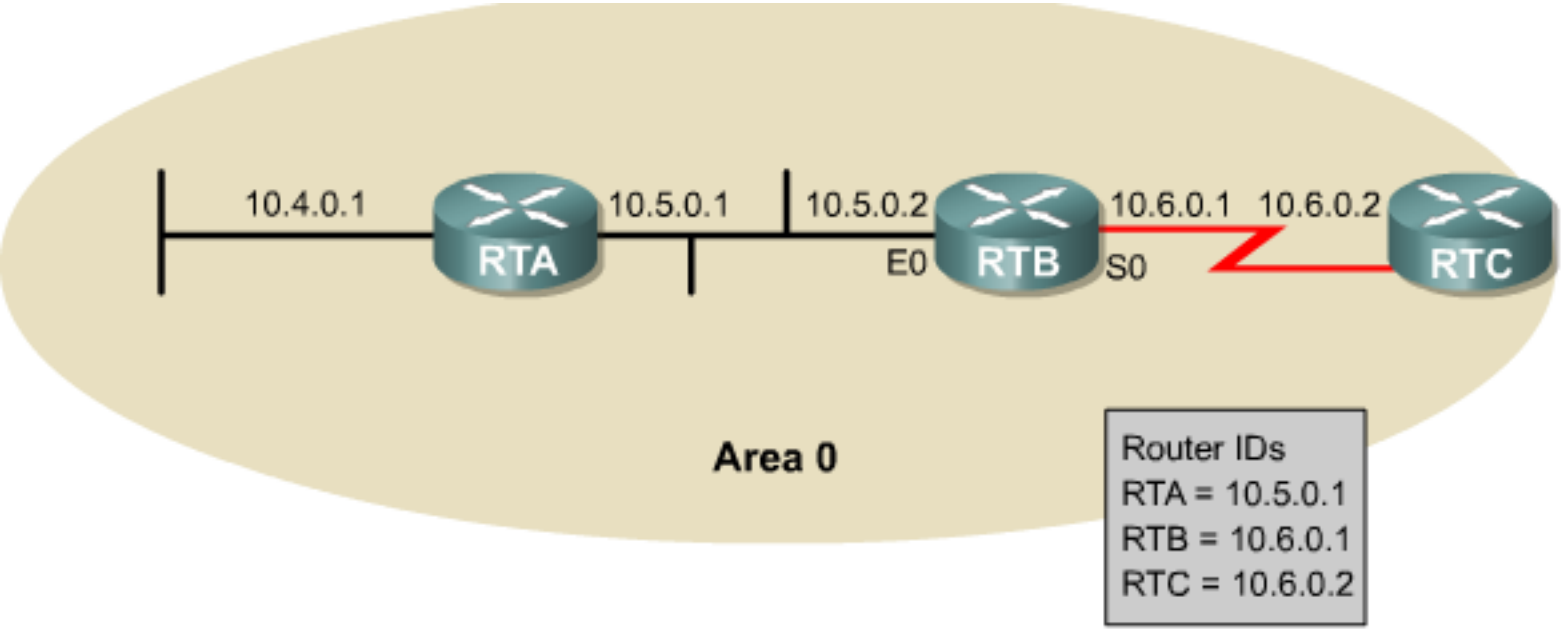
Steps in the Operation of OSPF

Discover neighbors



An OSPF router tries to form an adjacency with at least one neighbor for each IP network to which it is connected.

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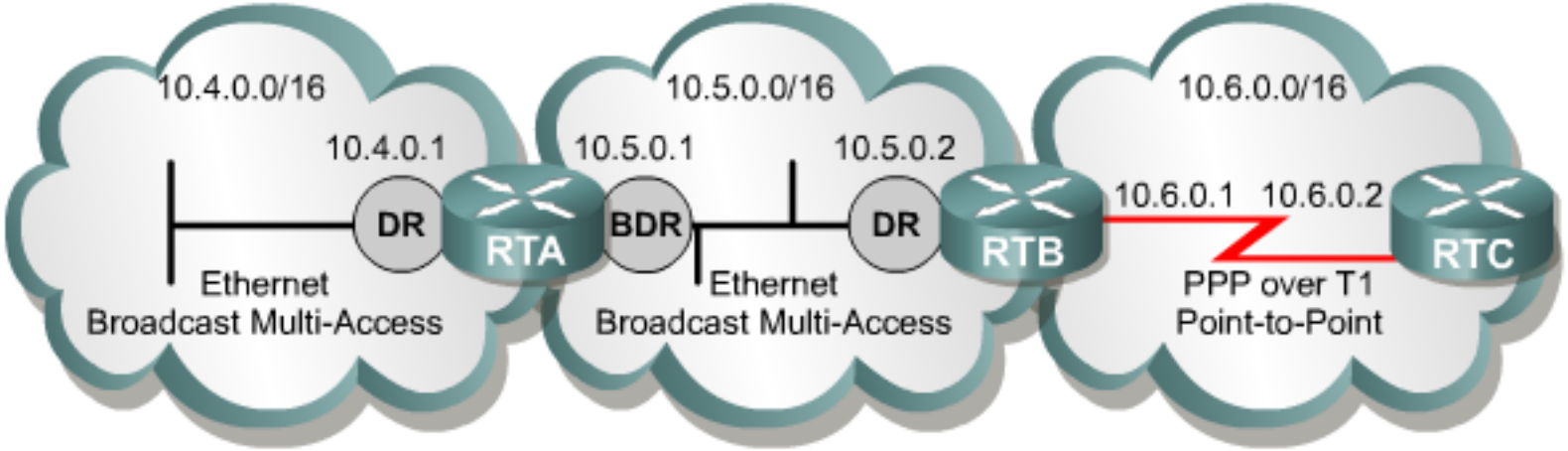
Steps in the Operation of OSPF

Elect DR and BDR on Multi Access Network



OSPF routers perform DR and BDR elections only on multiaccess IP networks.

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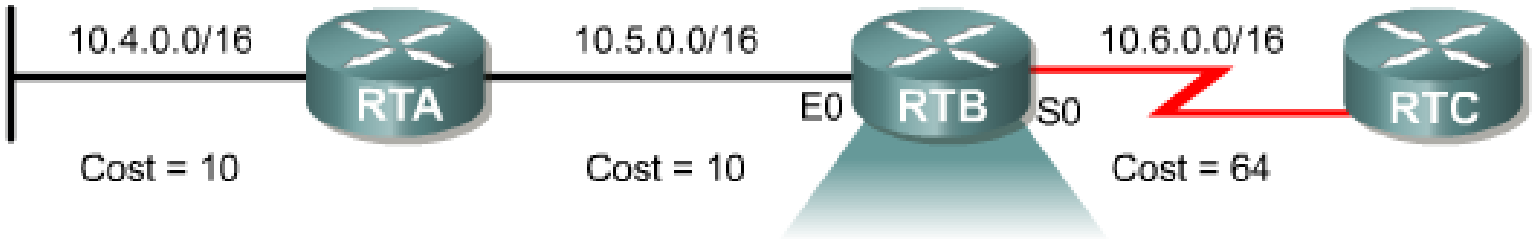
Steps in the Operation of OSPF

Selecting the Best Route



Net	Cost	Out Interface
10.4.0.0	20	E0

The link state database is processed by the shortest path first algorithm and the best routes selected.

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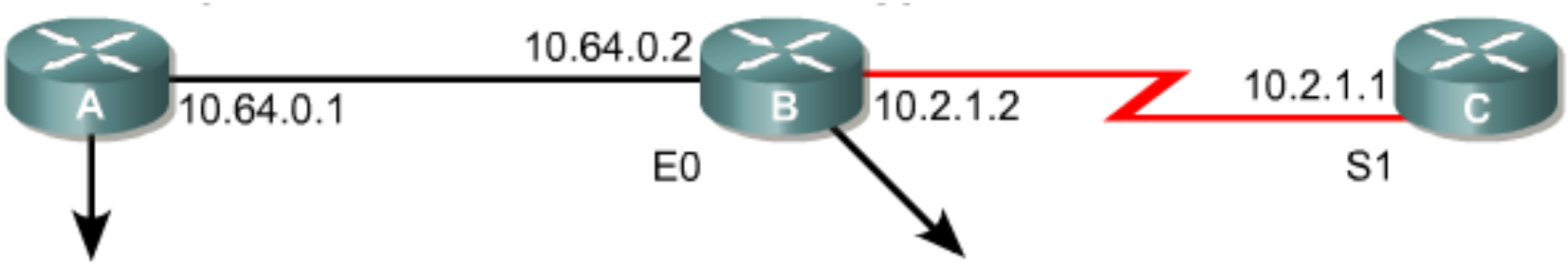
Basic OSPF Configuration




```
<Output Omitted>
interface Ethernet0
ip address 10.64.0.1 255.255.255.0
!
<Output Omitted>
router ospf 1
network 10.64.0.0 0.0.0.255 area 0
```

```
<Output Omitted>
interface Ethernet0
ip address 10.64.0.2 255.255.255.0
!
interface Serial0
ip address 10.2.1.2 255.255.255.0
<Output Omitted>
router ospf 1
network 10.2.1.2 0.0.0.0 area 0
network 10.64.0.2 0.0.0.0 area 0
```

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Basic OSPF Configuration

Network area Command	Description
address	Can be the network address, subnet, or the address of the interface. Instructs router to know which links to advertise, which links to listen to advertisements on, and what networks to advertise.
wildcard-mask	An inverse mask used to determine how to read the address. The mask has wildcard bits where 0 is a match and 1 is "do not care"; for example, 0.0.255.255 indicates a match in the first two bytes. (the equivalent REGULAR subnet mask would be a 16 bit mask of 255.255.0.0) If specifying the interface address, use mask 0.0.0.0.
area-id	Specifies the area to be associated with the address. Can be a number or can be similar to an IP address A.B.C.D. For a backbone area, the ID must equal 0.

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Configuring OSPF Loopback Address and Router Priority


```
! Create the loopback 0 interface
Sydney3(config)#interface loopback 0
Sydney3(config-if)#ip address 192.168.31.33
255.255.255.255
Sydney3(config-if)#exit
! Remove loopback 0 interface
Sydney3(config)#no interface loopback 0
Sydney3(config)#
01:47:27: %LINK-5-CHANGED: Interface Loopback0, changed
state to administratively down
```

A loopback is a software only interface. To remove a loopback interface enter **`no interface loopback`**.

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Setting OSPF Priority


```
Sydney1(config)#interface fastethernet 0/0
Sydney1(config-if)#ip ospf priority 50
Sydney1(config-if)#end
Sydney1#
00:21:57: %SYS-5-CONFIG_I: Configured from console
by console
```

The Hello packet sent on the fastethernet interface will have the Router Priority Field set to 50.

The priorities can be set to any value from 0 to 255. A value of 0 prevents that router from being elected. A router with the highest OSPF priority will win the election for DR.

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Modifying OSPF Cost Metric

Medium	Cost
56 kbps serial link	1785
T1 (1.544 Mbps serial link)	64
E1 (2.048 Mbps serial link)	48
4 Mbps Token Ring	25
Ethernet	10
16 Mbps Token Ring	6
100 Mbps Fast Ethernet, FDDI	1

```
Sydney2(config-if)#ip ospf cost ?
  <1-65535>  Cost
Sydney2(config-if)#ip ospf cost 1
```

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Configuring OSPF Authentication

Cisco

```
Sydney1(config-if)#ip ospf message-digest-key 1 md5 7
asecret
Sydney1(config-if)#exit
Sydney1(config)#router ospf 1
Sydney1(config-router)#area 0 authentication message-
digest
Sydney1(config-router)#end
Sydney1#
```

Interface and Router configuration is required.

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Configuring OSPF Timers

Cisco

```
Sydney1(config-if)#ip ospf hello-interval 5
Sydney1(config-if)#ip ospf dead-interval 20
```

OSPF timers are configured on the interface.

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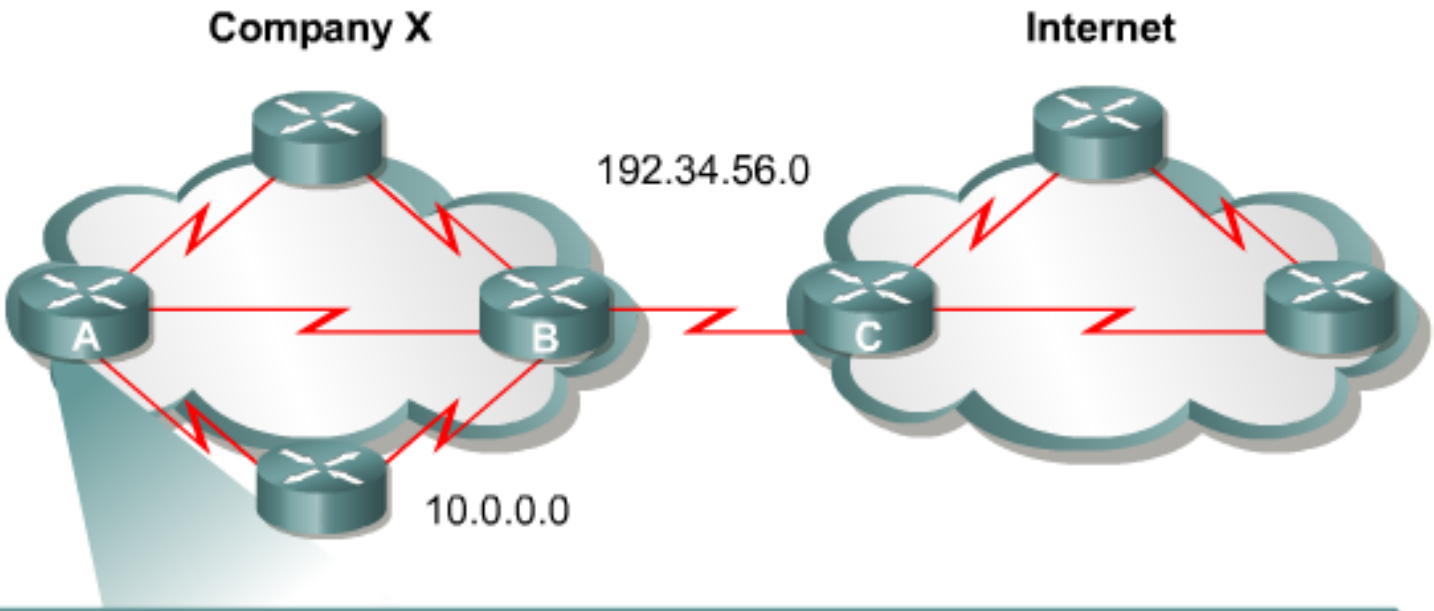
OSPF - Propagating a Default Route



Routing Table

No entry for destination net
Try router B default route

Use if next hop is not explicitly listed in routing table

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Common OSPF Configuration Issues

No Neighbor	OSPF Routes Not Shown
Do interfaces have same OSPF timers?	Do interfaces have correct IP address and subnet mask?
Do connected interfaces have same network type?	Do network statements have correct wildcard masks?
Are authentication keys and passwords the same on interfaces?	Do network statements put links into correct area?
Do the router neighbors have duplicate IP addresses?	
Is the router interface up?	

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Verifying OSPF Configuration

- `show ip protocol`
- `show ip route`
- `show ip ospf interface`
- `shop ip ospf`
- `show ip ospf neighbor detail`
- `show ip ospf database`

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The `debug` and `clear` Commands for OSPF Verification

Command	Description
`clear ip route *`	Clear all routes in routing table
`clear ip route a.b.c.d`	Clear route to a.b.c.d in routing table
`debug ip ospf events`	Report all OSPF events
`debug ip ospf adj`	Report OSPF adjacency events

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